

1. Requirements Table

1.1 Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow subscribers to swap monthly box items or pause/skip deliveries up to 48 hours prior to the monthly billing renewal date.	High	Pass: Selection updated and fulfillment manifest refreshed. Fail: Customization locked out prematurely or change not reflected in manifest.	Gives subscribers control over their box before fulfillment is locked in, reducing cancellations and unwanted charges.
FR-002	Functional	The system shall let subscribers customize their monthly box by selecting from item categories (e.g., books, coffee, snacks) filtered by their saved preference tags.	High	Pass: Box contents reflect only items matching selected/allowed preference tags. Fail: Items outside subscriber's preference tags are included without override.	Personalisation is the core value proposition of the subscription box service.
FR-003	Functional	The system shall allow a subscriber to skip a single billing cycle without cancelling the underlying subscription, provided the request is made before the swap/pause cutoff.	Medium	Pass: No box is fulfilled and no charge is generated for the skipped cycle; subscription auto-resumes next cycle. Fail: Subscriber is billed or a box is shipped for a skipped cycle.	Reduces churn by giving subscribers flexibility instead of forcing full cancellation.
FR-004	Functional	The system shall allow subscribers to view their past delivery history and billing history, including swapped/skipped cycles, from their account dashboard.	Medium	Pass: History list shows accurate past cycles with status (delivered/skipped/swapped) and matching invoice. Fail: History omits a cycle or shows incorrect	Supports transparency and reduces support tickets about billing/delivery disputes.
FR-005	Functional	The system shall allow the Fulfillment Lead to generate a fulfillment manifest that consolidates all finalized subscriber box selections after the customization window closes, and export shipping labels for the batch.	High	Pass: Manifest includes every finalized order with correct items and address; labels generated for 100% of manifest entries. Fail: Any finalized order is missing from the manifest or label generation fails silently.	Fulfillment operations depend on an accurate, complete manifest to pack and ship boxes on schedule.

1.2 Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The fulfillment manifest generator must export shipping labels for 10,000 monthly boxes in under 1 minute, over an encrypted connection.	High	Pass: Benchmarking tests confirm target latency and security standards (TLS 1.2+, authenticated access) under simulated peak load. Fail: Export exceeds 1 minute or data is transmitted unencrypted.	Fulfillment runs on a fixed monthly cutoff; slow exports delay shipping for the entire batch, and shipping/PII data must stay protected.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-002	Usability & Reliability	The subscriber-facing customization portal must remain available with 99.9% uptime during the 48-hour swap/pause window before each monthly renewal, and pages must load within 3 seconds on a standard 4G connection.	Medium	Pass: Uptime and load-time monitoring confirm targets are met during each renewal window across a full billing cycle. Fail: Portal is unavailable or unresponsive during any part of the swap window.	The swap/pause window is time-boxed and non-extendable, so downtime directly causes subscribers to lose the ability to customize or skip, driving complaints and churn.

2. Use-Case Flow Specification

Use Case: Swap Monthly Box Items / Pause Delivery Cycle

Primary Actor: Subscriber

Secondary Actor: none (Fulfillment manifest is updated automatically by the system)

Related Requirement: FR-001

Preconditions

- The Subscriber is registered and successfully authenticated on the Subscription Box Scheduler portal.
- The Subscriber has an active subscription with a box scheduled for the upcoming billing cycle.
- The current date/time is prior to the monthly billing renewal date.

Postconditions

- Success: The Subscriber's box selection (or pause/skip status) is updated, the fulfillment manifest entry is refreshed to match, and a confirmation is sent to the Subscriber.
- Failure: No changes are persisted; the Subscriber's original box selection and manifest entry remain unchanged.

Main Success Scenario

1. The Subscriber logs in to the portal (includes Authenticate User).
2. The Subscriber navigates to "My Subscription" and selects "Customize This Month's Box."
3. The system displays the current box items, the Subscriber's saved preference tags, and available substitute items or a pause/skip option.
4. The Subscriber selects replacement items to swap, or chooses to pause/skip the cycle, and confirms the change.
5. The system validates that the request falls within the 48-hour pre-renewal customization window and that any selected substitute items are in stock.
6. The system saves the updated box selection (or pause/skip status) against the Subscriber's account.
7. The system regenerates the Subscriber's entry in the fulfillment manifest to reflect the change.
8. The system sends a confirmation notification (email and in-app) to the Subscriber summarizing the update.
9. The use case ends successfully.

Alternate Flow — A1: Customization Window Closed

Trigger: At step 5 of the main flow, the system determines that fewer than 48 hours remain before the billing renewal date.

1. The system rejects the swap/pause request and displays the message: "Customization window closed for this cycle."
2. The system offers the Subscriber the option to apply the requested change to the following month's cycle instead.
3. The Subscriber either confirms the deferred change or cancels the request.

4. If deferred, the system schedules the change for the next billing cycle and sends a confirmation noting it will take effect next cycle; the current cycle's box and manifest entry remain unchanged. The use case then ends.
2. Use-Case Flow Specification

3. UML - DIAGRAM

